Study Protocol

Study title

Defining changes in nasal immunity that favour propensity for pneumococcal carriage in People Living with HIV

Short title

Nasal Immunity and Pneumococcal Colonisation

Reference details

National Research Ethics Committee
2023 -01- 24
APPROVED

Study reference:	NASOMUNE
NHSRC Reference:	21/24/2680
LSTMREC Reference:	21-035
Date and Version number:	3rd November 2022, v3.0
Principal Investigator:	Dr Kondwani Jambo
Sponsor:	Liverpool School of Tropical Medicine
Funding body:	MRC UK
Author:	Dr Kondwani Jambo and Dr Newton Kalata

Confidentiality Statement

This document contains confidential information that must not be disclosed to anyone otherthan the Sponsor, the Investigator Team, and members of the Research Ethics Committee, unless authorised to do so. This information cannot be used for any purpose other than the evaluation or conduct of the clinical investigation without the prior written consent of Dr Kondwani Jambo.

Contents

Study Protocol 1
Study title 1
Short title 1
Reference details 1
Confidentiality Statement 1
Statement of Compliance 5
Investigator Agreement 5
Conflict of Interest 5
Key study contacts and roles 6
Abstract 7
Schedule of visits and procedures 8
Abbreviations 9
Background 10
Problem statement 10
Justification 10
Study objectives 11
Main objective 11
Study Hypothesis 11
Literature review 11
Immunity against pneumococcal carriage in humans 11
Modulation of nasopharyngeal innate defences promote acquisition and establishment ofpneumococcal carriage 12
HIV infection impairs systemic and lower respiratory tract immunity 13
Methodology 14
Study Design 14
Study Endpoints 14
Recruitment Target 15
Duration 15
Method details 16
Participant recruitment 16
Inclusion criteria: 17
Exclusion criteria: 17
Sample and data collection 17
Clinical and demographic data 17
Nasal swab 18

National Health Sciences Research Committee
2023 -01- 24
APPROVED

Blood 18
Nasorportion strips 18
Nasal cells 18
Poke and coughs 19
Determination of colonisation 19
Immunological assays 19
Analysis plan and sample size justification 21

Safety considerations and assessment 22
Evaluation of Serious Adverse Events (SAE) 22
Risks to researchers 23
COVID-19 risk and how it will be mitigated 23

Data analysis 23
Mixed effect generalised linear model development 24

Dissemination of findings 24

Ethical considerations 24
Autonomy 24
Non-maleficence 24
Beneficence 25
Justice 25
Protection of vulnerable groups 25

Remuneration 26

Personnel roles and institutions 27

Confidentiality and anonymity 28

Samples and data 28
Sample and data storage 28
Collaborating laboratories 28

Sponsorship and Indemnity 29

Amendments 29

Capacity strengthening 29

Budget 30
Justification of the Budget 30

References 32

National Health Sciences Research Committee
2023 -01- 24
APPROVED

Statement of Compliance

Investigator Agreement

"I have read this protocol and agree to abide by all provisions set forth therein. I agree to comply with the International Conference on Harmonisation Tripartite Guideline on Good Clinical Practice."

Kondwani Jambo

Chief Investigator | Investigator's Signature | Date: 03rd November 2022

Conflict of Interest

"According to the Declaration of Helsinki, 2008, I have read this protocol, and declare no conflict of interest with any Investigators"

Kondwani Jambo

Chief Investigator | Investigator's Signature | Date: 03rd November 2022

National Health Sciences Research Committee
2023 -01- 24
APPROVED

Key study contacts and roles

Principal Investigator **Dr Kondwani Jambo** Group Head, Viral Immunology	Malawi-Liverpool-Wellcome Trust Clinical Research Programme, P.O Box 30096, Chichiri, Blantyre 3, Malawi Phone: +265 1 876444 Email: kjambo@mlw.mw
Collaborator (Genomics) **Dr Chrispin Chaguza** Postdoctoral Fellow	Wellcome Trust Sanger Institute, Cambridge, UK Email: cc19@sanger.ac.uk
Collaborator (Immunology) **Prof Daniela Ferreira** Professor of Respiratory Vaccines and Infection Immunology	Liverpool School of Tropical Medicine Pembroke Place, Liverpool L3 5QA UK. Emali: daniela.ferreira@lstmed.ac.uk
Collaborator **Dr Ken Malisita** Medical Director	Queen Elizabeth Central Hospital Lighthouse, Blantyre 3, Malawi Email: ekari988@gmail.com
Collaborator **Dr Chris Moxon** Clinician Scientist	University of Glasgow, UK Email: cmoxon@mlw.mw
Student (Immunology) **Joseph Phiri** PhD Student	Malawi-Liverpool-Wellcome Trust Clinical Research Programme,P.O Box 30096, Chichiri, Blantyre 3, Malawi Email: jphiri@mlw.mw
Student (Microbiology and Genomics) **Lusako Sibale** PhD Student	Malawi-Liverpool-Wellcome Trust Clinical Research Programme,P.O Box 30096, Chichiri, Blantyre 3, Malawi Email: lsibale@mlw.mw

National Health Sciences Research Committee
2023 -01- 24
APPROVED

Abstract

Study Title: Defining changes in nasal immunity that favour propensity for pneumococcal carriage in People Living with HIV

Study Identifier: NASOMUNE

Chief Investigator: Dr Kondwani Jambo

Study Centres: MLW, QECH Lighthouse and Gateway Clinic (Blantyre DHO)

Study Participants: Healthy adults aged 18-45 years; HIV-uninfected adults, People Living with HIV (PLHIV) on ART for <3months and PLHIV on ART for >1yr.

Planned study size: 195 participants (1:1:1)

Follow-up duration: 4 months

Sampling: See visit schedule

Study interventions: Nasal swab
Nasal washings, nasal absorption
Nasal cells by scraping (rhinoprobe)
Nose poke
Cough on a blood agar plate
Venipuncture

Study duration: estimated end December 2023

Planned start date: July 2021

Health Sciences Research Committee
2022 -01- 24
APPROVED

Study aims

Primary objective: To determine association between innate immune cells in the nasal mucosa and pneumococcal carriage dynamics in ART-experienced PLHIV compared to age-and sex-matched HIV-uninfected healthy controls.

Outcome measure: Correlation between innate immune cell frequency and function with pneumococcal carriage density and duration.

Secondary objectives: 1) To define the nasal cytokine milieu and transcriptomic signature associated with increased predisposition to pneumococcal carriage in ART-experienced PLHIV.
2) To determine the carriage acquisition rates between HIV-uninfected and ART-experienced PLHIV.
3) To determine the association between presence of viral or bacterial respiratory pathogens in the nasal mucosa to pneumococcal carriage dynamics
4) To explore the impact of pneumococcal carriage in ART-experienced PLHIV on household transmission dynamics.

Outcome measures: 1) Cytokine and gene signatures associated with propensity for pneumococcal carriage.
2) Number carriage episodes in HIV-uninfected and ART-experienced PLHIV.
3) Identification of respiratory pathogens associated with increased pneumococcal carriage density.
4) Frequency of pneumococcal carriage caused by the same serotype

Allocation method 1:1:1, Convenience sampling

Schedule of visits and procedures

Study Visit	A	B	C	D	E	F	G		K	L	M	N	O	Q
Study Week	**1**	**2**	**3**	**4**	**5**	**6**	**7**		**12**	**13**	**14**	**15**	**16**	**17**
Consent (Written)	x													
Consent (Verbal)	x	x	x	x		x	x	x	x	x	x	x	x	x
Medical History	x													
HIV test	x													x
FBC (2ml)	x													x
Nasal swab	x	x	x	x		x	x	x	x	x	x	x	x	x
Serum (5ml)	x						x		x					x
PBMCs (20ml)	x						x		x					x
Nasal scrape	x	x	x	x		x								
Nasosorption	x	x	x	x		x								
Poke	x	x	x	x		x								
Cough	x	x	x	x		x								
Household Sampling														
Nasal swab	x	x	x											

Week 8 to 11

Table 1: Study procedures and sampling schedule. There will be a tolerance of ± two working days for all study visits

National Health Sciences Research Committee
2023 -01- 24
APPROVED

Abbreviations

ART	Antiretroviral Therapy
CRF	Case Report Form
ELISPOT	Enzyme-Linked Immune Absorbent Spot
GCP	Good Clinical Practice
HIV	Human Immunodeficiency Virus
HTS	Human immunodeficiency virus Testing Service
ICH-GCP	International Conference on Harmonisation of Good Clinical Practice
LSTM	Liverpool School of Tropical Medicine
MK	Malawian Kwacha
MLW	Malawi-Liverpool Wellcome Trust Clinical Research Programme Laboratories
MTA	Material Transfer Agreement
NHSRC	National Health Sciences Research Committee
PBMC	Peripheral Blood Mononuclear Cell
PCV	Pneumococcal conjugate vaccines
PCV13	Pneumococcal conjugate vaccine 13
PLHIV	People Living With HIV
RNA	Ribonucleic Acid
SOP	Standardized Operating Procedure

National Health Sciences Research Committee
2023 -01- 24
APPROVED

Background

Streptococcus pneumoniae, also known as pneumococcus, is the most common bacterial cause of pneumonia, meningitis and otitis media worldwide (1). Despite its pathogenicity and associated antimicrobial resistance, the pneumococcus commonly colonises the nasopharynx in adults living in sub-Saharan Africa (10-60%) (2-4). Paradoxically, however, People Living with HIV (PLHIV) on antiretroviral therapy (ART) have higher rates of pneumococcal carriage (40-60%) than HIV-uninfected adults (8-15%) or PLHIV not on ART (18-25%) (3, 4).

Problem statement

Pneumococcal conjugate vaccine (PCV13) has been shown to reduce vaccine-type carriage and disease in most high-income countries, as well as, provide herd immunity to unvaccinated populations (5, 6). However, surveillance data from Malawi demonstrates persistently high pneumococcal carriage among PLHIV on ART despite the introduction of pneumococcal conjugate vaccine (PCV)13 vaccine in 2011, implying that the herd immunity benefits of vaccination are not robust in this setting. Due to the current test and treat strategy, the number of PLHIV on ART will expand tremendously, and in parallel, an increasein size of the pneumococcal transmission reservoir, this threatens the public health benefits of pneumococcal vaccination.

Justification

Pneumococcal carriage is a prerequisite for pneumococcal disease and is essential for transmission in the community, hence control of pneumococcal carriage is a key strategy for preventing pneumococcal disease. This underscores the need for a better understanding of the mechanism underlying susceptibility to pneumococcal carriage in PLHIV on ART, which is central to development of new approaches to maximise the public health benefit of pneumococcal vaccination in high transmission settings.

National Health Sciences Research Committee
2023 -01- 24
APPROVED

Study objectives

Main objective

To determine association between innate immune cells in the nasal mucosa and pneumococcal carriage dynamics in ART-experienced PLHIV compared to age-and sex-matched HIV-uninfected healthy controls.

Specific objectives

a. To define the nasal cytokine milieu and transcriptomic signature associated with increased predisposition to pneumococcal carriage in ART-experienced PLHIV.
b. To determine the carriage acquisition rates between HIV-uninfected and ART-experienced PLHIV.
c. To determine the association between presence of viral or bacterial respiratory pathogens in the nasal mucosa to pneumococcal carriage dynamics
d. To explore the impact of pneumococcal carriage in ART-experienced PLHIV on household transmission dynamics.

Study Hypothesis

Altered innate immune cell phenotype and function in the nasal mucosa is associated with reduced clearance of pneumococcal carriage leading to high carriage density and longer duration of carriage in PLHIV on antiretroviral therapy.

National Health Sciences Research Committee
2023 -01- 24
APPROVED

Literature review

Immunity against pneumococcal carriage in humans

There are substantial data from humans and animal models describing the potential adaptive immune mechanisms that are required for protection against pneumococcal acquisition or clearance of carriage (7). The current working model in humans suggests that high levels of circulating memory B cells to capsular polysaccharide associate with protection against carriage acquisition. Quick differentiation of memory B cells into plasma cells leads to antibody production. Plasma cells produce pneumococcal serotype- and protein-specific antibodies, which function by agglutination and opsonisation, resulting in enhanced mucociliary clearance and phagocytosis, respectively (7-9). Furthermore, interleukin-17A (IL-

National Health Sciences Research Committee
2023 -01- 24
APPROVED

Methodology

Study Design

A prospective longitudinal cohort study.

This study is based on the foundation of our COMREC-approved PNEUMODUDE study(P.11/18/2532) and will utilise standard operating procedures successfully validated in Malawi.

The study will be carried out in three stages:

1. Screening and recruitment: Potential participants will be screened to ensure health and safety as detailed in the methods section.
2. Detection of pneumococcal carriage and other respiratory pathogens: Nasal swab samples will be taken weekly for entire duration of the study, according to a standardized protocol. In addition, a cough and poke sample will be collected weekly in the first 28 days of the study. Classical microbiological culture will determine nasal colonisation andpneumococcal serotypes at each time point in all the three sample types. We will also usemolecular techniques to determine the presence of other respiratory pathogens as they are potential confounders to the study.
3. Immunology response measurements: Blood cells, nasal cells and nasal lining fluid samples will be taken to determine the immunological response to pneumococcal carriage.

Study Endpoints

Primary endpoint: Correlation between innate immune cell frequency and function with pneumcoccoccal carriage density and duration.

Secondary endpoints: Cytokine and gene signatures associated with propensity for pneumococcal carriage. Number carriage episodes in HIV-uninfected and ART-experienced PLHIV. Identification of respiratory pathogens associated with increased pneumococcal carriage density. Frequency of pneumococcal carriage caused by the same serotype

Study Setting: Clinical procedures will be conducted at the Queen Elizabeth Central Hospital, Lighthouse, Blantyre, Malawi. Laboratory procedures will be conducted at the adjacent Malawi-Liverpool Wellcome Trust (MLW) Clinical Research Programme Laboratories.

Recruitment Target

We will recruit 126 asymptomatic adult participants to complete the study. PLHIV on ART will be recruited at the QECH Lighthouse, while age- and sex-matched HIV-infected controls will be recruited at the Gateway Clinic (Blantyre DHO).

Duration

Recruitment of all participants and follow up will be completed within 24 months. A Gantt chart demonstrates study activities (Figure 2).

Figure 2: Gantt Chart

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Year 1 (2021)												
Ethical approval												
Recruitment of study team, set up and trainings												
Screening, enrolment and follow up of participants, lab results												
Year 2 (2022)												
Screening, enrolment and follow up of participants, lab results												
Lab analysis-sequencing and RNA transcriptomics, Cytokines												
Data analysis and publication												
Year 3 (2023)												
Lab analysis-sequencing and RNA transcriptomics, Cytokines												
Final data analysis and publication												

National Health Sciences Research Committee
2023 -01- 24
APPROVED

National Health Sciences Research Committee
2023 -01- 24
APPROVED

Method details

Participant recruitment

PLHIV on ART will be approached by the study nurses at the QECH Lighthouse for participation into the study. Individuals who show interest will be handed the participant information leaflet and a research team member will explain the study to the potential Participant. The potential participant will be allowed to ask questions to the study recruitment team. Potential participants willing to give consent the same day will be enrolled after giving informed consent. Those that would like more time to think before consenting will be given the participant information leaflet which has contacts of the study team. They will be free to contact the team once they are willing to give informed consent at any time within the study enrolment period. Age- and sex-matched HIV-uninfected controls will be recruited at the Gateway Clinic in a similar manner to the PLHIV stated above. In a subset of index participants positive for pneumococcal carriage and willing to have members of theirhousehold enrolled to the study, we will arrange a specific day for the members of their household to meet the team for informed consent or assent (if less than 18 years old). The index participant will provide contact details for adult household members to the study team, who will approach the members by mobile communication and invite them to come on an agreed day for a formal informed consent process. Children will be invited through their parent or guardian (who may either be the index participant or the adult household member). Informed consent will take place in the clinic, with each member to ensure confidentiality.

Informed consent process for both index participant and household members will be conducted by either a research nurse or research doctorswho are formally delegated by the Chief Investigator, trained in GCP, consent and the trial protocol. The nurse or doctor will explain the participant information leaflet to the potential index participant or household member, giving them time to ask questions. The research team will also have time to ask potential participants questions to check their comprehension of the study. The whole consenting process should take a minimum of 30 minutes. Once the participant and study team are satisfied, they will both sign an informed consent form in the presence of an impartial witness if appropriate. Following consent, inclusion and exclusion criteria will be applied as below.

Inclusion criteria:

- Asymptomatic adults aged 18-45years who are HIV-uninfected or PLHIV on ART and have at least one under five child in the household. Under-5 children have the highest pneumococcal carriage prevalence and are the main source of transmission in the household. Therefore, their presence will increase the likelihood of identifying a pneumococcal carriage positive adult participant.
- Household member of *index participants.

  *An index participant is a participant who is either pneumococcal carriage positive or pneumococcal carriage negative.

Exclusion criteria:

- Acute illness
  - Current illness
  - Acute illness within 3 days prior to enrolment
  - Antibiotic use apart from co-trimoxazole prophylaxis in the last 4 weeks
- Taking immunosuppressive medication (apart from ARVs) that may include but is not limited to steroids and steroid nasal spray.
- History of Smoking
  - Current regular smoker (smokes daily/ smokes > 5 cigarettes per week)
  - Recent smoker i.e. within the last 6 month
  - Ex-smoker with a significant smoking history (>10 pack years)
- Unable to give informed consent
- Known or suspected pregnancy
- Chronic respiratory illness

National Health Sciences Research Committee
2023 -01- 24
APPROVED

Sample and data collection

Following consent, inclusion and exclusion criteria, data and samples will be collected according table 1 above.

Clinical and demographic data

We will use electronic case report forms (eCRF) to collect demographic and clinical data,

including age, sex, antibiotic use, HIV status, household characteristics, geographic location. For the index, the following eCRFs will be used; Screening (Appendix 5), Enrollment (Appendix 6), Follow up (Appendix 7) and Termination (Appendix 8). While for the household contact, the following eCRFs will be used; Enrollment(Appendix 9), Follow up (Appendix 10) and Termination (Appendix 11).

Nasal swab

NS will be performed using a standard protocol. This is a well-used and validatedtechnique to collect nasal bacterial specimens used successfully during the PNEUMODUDE study. Briefly, using an aseptic technique, the sterile swab is inserted into the nostril along the nasal septum just above the floor of the nasal passage to the nasopharynx, until a resistance is felt. Then it is held at the nasopharynx for upto 3-5 seconds gently rotating 360 degrees then removed and placed in a cryovial with 1.5ml of transport medium.

Research Committee
2023 -01- 24
APPROVED

Blood

Blood samples will be obtained using venipuncture by an appropriately trained team member. The volume of blood collected for each laboratory test isdetailed in Table 1.

Nasorportion strips

Nasosorption will be obtained before nasal scrapes and nasal swab. Thestrips are similar to blotting paper and developed by Hunts Development Ltd (UK). Strips collect concentrated nasal lining fluid before the nasal wash to measure inflammatory responses induced by infection that may be associated with increased colonisation density and acquisition. Concentrated nasal fluid will be used to measure cytokine levels by multiplexbead array. Blotting paper will be held inside the nostril for up to 3 minutes until soaked.

These will then be removed and placed in a microcentrifuge tube for storage.

Nasal cells

Cells will be collected using a nanosampling method in which cells are obtained through minimally invasive superficial nasal scrape biopsies (rhinoprobe). Experience from our feasibility study (29) demonstrates that participants can be biopsied multiple times with no significant side effects. Nasal cell samples will be obtained after nasal wash samples have been

taken. Up to 4 samples (2 per nostril) will be obtained at each nasal sampling visit. If the researcher finds that the sample is insufficient, for example no cells are visible on the rhinoprobe, the sample can be repeated immediately. The rhinoprobes are placed in transport media immediately after collection.

Poke and coughs

Participants will be asked to poke their nose using their finger and directly coughing onto an Blood agar plate (30). This method has been successfully validated and implemented in Malawi using our PNEUMODUDE study (COMREC Number P.11/18/2532). It is minimally invasive and highly tolerable by participants. After poking their nose, the finger is swabbed with a cotton bud and the spacemen is stored in transport media. Both pock and cough will allow us to assess two different methods of pneumococcal shedding as surrogates for transmission.

Determination of colonisation

Colonisation will be defined by the microbiology result of nasal swabs at any time point. Nasal swabs will be processed and plated on to culture media and incubated overnight at 37°C in 5% carbon dioxide (CO_2). Colonies will be confirmed as *S. pneumoniae* using classical microbiological techniques including (i) typical draughtsman-like colony morphology, (ii) the presence of α-haemolysis, (iii) optochin sensitivity and (iv) bile solubility. Typing by latex agglutination will be done using a commercial kit to confirm pneumococcal serogroup.
Isolates will be frozen at -80°C for storage. Monitoring of colonisation will include measurement of pneumococcal density by LytA PCR ascertainment of duration of carriage using sequencing (to ascertain the isolate overtime).

National Health Sciences Research Committee
2023 -01- 24
APPROVED

Immunological assays

- Cellular responses: Measurement of frequency and function of innate immune cells will be performed on peripheral blood and nasal samples using flow cytometry and ELISPOT. This will allow us to identify potential cellular correlates of protection againstpneumococcal colonisation.
- Cytokine profiles: Measurement of cytokines will be done on nasosorption and serum samples using multiplex bead array. This will enable us to identify potential soluble

marker correlates of protection against pneumococcal colonisation.

- Host transcriptomic analysis: Measurement of host transcriptomic profile will be done using RNA sequencing (RNAseq) on nasal cells and peripheral blood. This will allow us to characterise potential nasal or blood transcriptomic signatures associated with protection or susceptibility to pneumococcal carriage.

National Health Sciences Research Committee
2023 -01- 24
APPROVED

Analysis plan and sample size justification

The primary endpoint is the correlation between innate immune cell frequency and function with pneumcoccoccal carriage density and duration. The secondary endpoints include cytokine and gene signatures associated with propensity for pneumococcal carriage, numbercarriage episodes in HIV-uninfected and ART-experienced PLHIV. Identification of respiratory pathogens associated with increased pneumococcal carriage density and frequency of pneumococcal carriage caused by the same serotype.

Studies of this nature are very challenging to calculate sample size. However, our experimental human pneumococcal carriage feasibility study (36) showed that failure to recruit immune cells into the nasal mucosa increases the likelihood of acquiring pneumococcal carriage (Non carriers vs. carriers; 75000cells/scrape vs. 47500 cells/scrape; standard deviation 28,452). We have used this data and assumed that PLHIV on ART will havereduced number of innate immune cells in the nasal mucosa than HIV-uninfected adults assuming equal variability of immune cell recruitment between PLHIV on ART<3 months and PLHIV on ART>1 year. Based on these parameters, we will need to follow up 32 adults in each group (total of 96)for power of 90% at significance level of 1%. We expect a 30% loss to follow up, hence we will follow up a total of 126 adults (42 in each group). The follow-up period is based on our preliminary data from the PNEUMODUDE study (COMREC Number P.11/18/2532) that has shown that the median pneumococcal carriage duration in adults in our setting is 43 days (95% CI; 41.5 to 45.5), hence within four months it would allow us to observe at least 2 carriage episodes per individual.

To evaluate transcriptional landscape in pneumococcal carriage positive participants and how it affects pneumococcal carriage dynamics (density and duration), we will recruit an additional number of index participants. Due to the exploratory nature of the study, no pilot data on nasal transcriptome in PLHIV is available to inform sample size calculation. However, previous studies have used an absolute fold change of >1.5 and a false-discovery rate (FDR) of <0.1 to identify differentially expressed genes. Assuming the same genes will be differentially expressed in PLHIV on ART, we will need to recruit an additional 18 adults in each group (total of 54) for a power of 90% at a significance level of 1%. Expecting a 30% loss to follow up, we

will follow up a total of 69 adults (23 adults in each group).

Safety considerations and assessment

Evaluation of Serious Adverse Events (SAE)

Any SAE as defined in ICH-GCP occurring to a research participant will be reported to NHSRC and study sponsor within 24 hours of the study team becoming aware of the event. In this event, the research will be stopped temporarily for investigation and any further work deferred until NHSRC advice.

Risks to researchers

These are standard clinical (needle stick) and laboratory (biohazard) risks. Experienced staff will carry out procedures that are within their competencies in accordance with standard operating procedures regulated by good clinical practice and national guidelines. Appropriate risk assessments are in place for all laboratory SOPs. All laboratory work will be conducted in an appropriately rated laboratory in line with health and safety regulations for research with human tissues / infectious agents.

COVID-19 risk and how it will be mitigated

Our study will collect aerosol generating samples which could pose a high risk for SARS-CoV-2 transmission to our research team. As such, research activities will follow the institutional and Malawi COVID-19 guidelines for health research that aim to minimise the risk of transmission in health research settings. In brief, study team will be provided with full personal protective equipment, our clinics will have access to hand washing facilities and social distance will be maintained. Study participants will be asked to wear facemasks before and after sample collection.

Data analysis

Exploratory analysis – we will employ simple descriptive statistics to estimate and display graphically the prevalence of pneumococcal carriage, carriage density, prevalence of multiple serotypes, frequency of cells and cytokine levels in each of the participant groups at baseline and at each follow-up time. Descriptive statistics will be reported as numbers and proportions for categorical variables; and median and range for continuous data. Categorical variables will be compared using χ^2 test, and continuous variables will be compared using thet-test or Mann-Whitney test. Statistical analysis and graphical presentation were performed

National Health Sciences Research Committee
2023 -01- 24
APPROVED

using GraphPad Prism (GraphPad Software, USA) or R statistical software (www.r-project.org).

Mixed effect generalised linear model development

Analysis of changes in carriage dynamics over time will use a mixed effect generalised linear model. For the model, the outcome measure will be pneumococcal carriage dynamics, calculated as area under the curve, a composite of carriage duration and carriage density. To protect against confounding with secular changes in carriage rate, the model will also include adjustments for seasonality, household structure, age and sex. The models will be fitted in the open-source R software environment (www.r- project.org) to enable reproducibility of all published results by researchers who do not haveaccess to proprietary software.

Dissemination of findings

The findings from this study will be disseminated amongst the scientific community. We intend to publish our findings in peer reviewed scientific journals and present data at appropriate local, national and international conferences. We will produce a close-out report for the NHSRC at the end of the study and a final report once data are published. In addition, we will produce a lay report of our findings which will be made available to all participants.

Ethical considerations

Autonomy

The participants will be given high-quality information that is written and spoken using common lay terms without jargon to allow them to understand the research objectives and the risks and benefits of all procedures. They will then be given time to consider the information before consenting to any involvement. At no stage should the participant feel pressured or persuaded into participating in the research study. Participants have the right to withdraw their consent and therefore withdraw from the study at any time without giving reason. If a participant withdraws from the study, we will recruit an alternative participant to complete the study.

National Health Sciences Research Committee
2023 -01- 24
APPROVED

Non-maleficence

Collection of samples will be as per the established and safe protocols and will be performed

by highly trained staff. Specific inclusion and exclusion criteria are set to further protect the participant. Experience and trained research staff will perform venepuncture and nasal sampling.

Beneficence

Participants will receive a health check including HIV-infection status assessment when taking part in this study. In addition, participants may benefit from a better understanding of clinical research, they may also benefit from a sense of contributing to medical research in a valuable way. Participants will be remunerated for their time and inconvenience – discussed below.

Justice

This must be balanced with non-maleficence. Inclusion and exclusion criteria are in place primarily to protect individuals from undue risk, as well as account for confounding factors to allow robust conclusions to be made from the data generated from the precious samples.

Protection of vulnerable groups

Our study will recruit children of any age that are household members of index participants who become positive for pneumococcal carriage while in the study. Their enrolment will help us to explore the impact of pneumococcal dynamics in ART-experienced PLHIV on household transmission dynamics (specific objective 4). All children will be asked to be involved in the study in the presence of their parent or guardian. Parental or guardian informed consent will be sought before approaching the child. Only children of parents or guardians who provide consent will be enrolled. Assent will be sought from children aged 8 to 17 years and this will have done in the presence of parent or guardian. For children aged below 8 years, only parent or guardian consent will be sought. Only a nasopharyngeal swab will be collected from household members of index participants, including children, according to the sampling schedule in Table 1 above. Well trained study nurses will be responsible for collecting swabs from children.

National Health Sciences Research Committee
2023 -01- 24
APPROVED

Remuneration

MARVELS Research Participant Remuneration (based on Malawian guidelines (37))			
Reimburse expenses	**Rate in MK**	**Number of events**	**Total**
a) Transport	900	16	14,400
b) Subsistence (one meal)	1000	17	17,000
Compensation			
Total time travelling (hrs)		16	
Total time in research facility (hrs)		8	
Time in days (day = 8 hours)	1000	16	11,600
Procedure A (mild discomfort)	2000	16	32,000
Procedure B (moderate discomfort)	6000	5	30,000
Procedure C (long or complex)	10000		
TOTAL for study			120,000
AVERAGE per visit			MK 7,500

Table 2: Research participant remuneration.

Venipuncture or nasal swab will cause mild discomfort.A combination of nasal scrape and nasal swab will cause mild/moderate discomfort.

It is intended that financial factors will not significantly influence an individual's decision to participate in this study. We reimburse participants for out of pocket expenses such as travel, and compensate time spent and burden. The sums offered in this study are consistent with remuneration guidelines published in Malawi, paid pro-rata (per activity and not dependent on study completion) (37).

If a participant withdraws from the study early, they will be compensated for the parts they took part in up until the time they withdrew. Payments are summarized in table 3, MK7,500 will be paid at the end of each visit. If the household members of the index participant volunteer to participate in the study, MK7,500 will be paid at the end of each study visit.

National Health Sciences Research Committee
2023 -01- 24
APPROVED

Personnel roles and institutions

Principal Investigator **Dr Kondwani Jambo** Group Head, Viral Immunology	Malawi-Liverpool-Wellcome Trust Clinical Research Programme, P.O Box 30096, Chichiri, Blantyre 3, Malawi Phone: +265 1 876444 Email: kjambo@mlw.mw
Collaborator (Genomics) **Dr Chrispin Chaguza** Postdoctoral Fellow	Wellcome Trust Sanger Institute, Cambridge, UK Email: cc19@sanger.ac.uk
Collaborator (Immunology) **Prof Daniela Ferreira** Professor of Respiratory Vaccines and Infection Immunology	Liverpool School of Tropical Medicine Pembroke Place, Liverpool L3 5QA UK. Emali: daniela.ferreira@lstmed.ac.uk
Collaborator **Dr Ken Malisita** Medical Director	Queen Elizabeth Central Hospital Lighthouse, Blantyre 3, Malawi Email: ekari988@gmail.com
Collaborator **Dr Chris Moxon** Clinician Scientist	University of Glasgow, UK Email: cmoxon@mlw.mw
Student (Immunology) **Joseph Phiri** PhD Student	Malawi-Liverpool-Wellcome Trust Clinical Research Programme,P.O Box 30096, Chichiri, Blantyre 3, Malawi Email: jphiri@mlw.mw
Student (Microbiology and Genomics) **Lusako Sibale** PhD Student	Malawi-Liverpool-Wellcome Trust Clinical Research Programme,P.O Box 30096, Chichiri, Blantyre 3, Malawi Email: lsibale@mlw.mw

National Health Sciences Research Committee
2023 -01- 24
APPROVED

Confidentiality and anonymity

Only authorised members of the research team will have access to any personal information. Only information of direct relevance to the study will be collected. All electronic records containing personal information will be stored in a password protected database on a password protected server at MLW. Electronic case report form data will be collected on encrypted study-specific tablet devices and synchronised daily onto the MLW server. This approach is well established at MLW, and the policy governing MLW data management policy is available upon request. Paper documentation containing personal information will be kept in a locked filing cabinet in a locked room in the Queen Elizabeth Central Hospital research clinic.

Each participant will be assigned a unique non-identifiable study number by a member of the clinical research team at recruitment. Unlinked non-identifiable clinical data will be stored and analysed at the MLW-laboratories and collaborating laboratories. Participant unlinked non-identifiable sequencing data will be made publicly available following international standards for data sharing and open science.

Samples and data

Sample and data storage

MLW will act as custodian for all data and samples collected during the study. Consent will be obtained from the participant to use the samples for this research only. Samples will be stored for a maximum of five years.

National Health Sciences Research Committee
2023 -01- 24
APPROVED

Collaborating laboratories

Samples will also be sent to national and international collaborating laboratories (MTA attached with application) to utilise specialist expertise not available in Malawi. Samples will be labelled with the anonymised study number. Consent will specifically be obtained from the participants to allow samples to be sent to our collaborators (Prof Ferreira at LSTM, UK; Dr Chaguza at Wellcome Sanger Institute, UK; Dr Chris Moxon at University of Glasgow, UK).